

AdMu - System Design Documentation

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Company: Seitai Michibiki (整体院導)

1. System Overview

AdMu is a B2B Software-as-a-Service (SaaS) platform designed specifically for local chiropractic and massage clinics in Japan. The system simplifies Google Ads management by automatically allocating budgets based on AI optimization and visualizing daily ad performance on a user-friendly dashboard.

2. Architecture & Data Flow

Technology Stack

- **Frontend:** Vanilla JS Single Page Application (SPA), HTML5, CSS3
- **Backend:** Python / FastAPI
- **Database:** PostgreSQL
- **Infrastructure:** Render.com Cloud Hosting

API Integration Data Flow

1. User securely inputs OAuth credentials on the Frontend UI.
2. Backend encrypts and stores credentials in PostgreSQL.
3. Once daily, or upon manual refresh, the Backend queries the Google Ads API via the official Python SDK.
4. The retrieved metrics (impressions, clicks, conversions) are formatted and served to the Frontend via a REST endpoint.

3. Required Google Ads API Features & Endpoints

AdMu specifically relies on the following Google Ads API services:

- **GoogleAdsService.search:** To fetch campaign metrics (Cost, Impressions, Clicks, Conversions) over a specified date range via GAQL.
- **CampaignService.mutate_campaigns:** To allow clinic owners to easily toggle campaign status (PAUSED / ENABLED) and adjust daily budgets through our simplified UI without needing to log directly into the complex Google Ads interface.

4. User Scenarios

Scenario 1: Daily Reporting

A clinic owner logs into AdMu in the morning. The dashboard immediately makes a call to our backend, which in turn queries the Google Ads API to display yesterday's cost and conversion data on a localized chart.

Scenario 2: Campaign Adjustment

If an ad campaign is underperforming based on our AI analysis, the user clicks "Pause Campaign" on our dashboard. The backend translates this action into a `CampaignOperation` and sends it via the API to pause the specific Google Ads campaign in real-time.